

TABLE 3: REARRANGED DESIGN MATRIX

FR		DP	Con- troller	Relay Spec- ifica- tion	Brake Relay Driver	Dum- my Load Relay Driver	Ambi- ent Wea- ther Sen- sor	Dead CPU Alarm Delay	Env. Norm	La- bour Prac- tice	Stor- age Spec- ifica- tion	Stor- age Mod- ule	NTC Sen- sor	Over- heat Alarm	Enclo- sure NEMA Std	PCB De- sign	En- clo- sure Ma- terial	En- clo- sure Col- our	Con- sole De- sign	Hall Effect Cur- rent Sen- sor	Wi-fi Mod- ule	Con- nec- tion Status Indica- tor	Org for After Sales	Org for Cus- tomer Sup- port	En- clo- sure Size	User Man- ual	Pow- er Sup- ply De- sign
	CODE	DP2	DP1	DP14	DP15	DP5	DP16	DP23	DP24	DP7	DP6	DP9	DP17	DP3	DP8	DP10	DP11	DP12	DP4	DP13	DP18	DP19	DP20	DP21	DP22	DP25	
Controller volt	M3	X																									
Alternator current sense	M5	X																									
Discharge current sense	M6	X																									
Batt. voltage sense	M7	X																									
Alternator voltage sense	M9	X																									
Operating volt	M1		X																								
Max current	M2		X																								
Safety margin	T2		X																								
Activate break circuit when dummy load is missing	S4			X																							
If dummy load is shorted then activate brake	S6			X																							
When wind speed exceeds a set threshold activate brake	S7			X																							
If battery absent, connect turbine to dummy load	S3				X																						
If battery shorted, connect power to dummy load	S5				X																						
By default connect turbine to dummy load	S8				X																						
Temp sense	M10					X																					
Humidity sense	M11					X																					
Test software functioning at regular interval	S9						X																				
Check CPU function and sound alarm if abnormal condition detected	T6						X																				
Parts environmental norm	T3							X																			
Sustainability norm	T4							X																			
Labour norm	B1								X																		
Data storage size	M13									X																	
Data storage technology	M12									X	X																
Sense temp of terminal block	M15											X															
Alarm if terminals over heat	S10												X	X													
Weather protection	M4													X													
Isolate connecting terminals from main controller	M14														X												
Provide heat and fire resistant material	T1															X											
Box colour	T2																X										
User interface	S1																	X									
Batt. Current sense	M8																		X								
Connectivity	S2																			X							
Raise warning from server/controller if connection fails	S11																				X						
After sales service	G1																					X					
Help line	G2																						X				
Device size	T1																							X			
User manual	G3																								X		
Alternate power source for circuit	T5																								X		

making things more convenient for us.

The steps of axiomatic design are:

1. Determine the CR
2. Describe the FRs and DPs for the conventional design process

3. Construct the design matrix
4. Rearrange the design matrix rows and columns to make minimum coupling
5. Find out the coupling source (if there is one)
6. Innovate to provide the solu-

tion for decoupling the matrix

7. Determine the FRs, DPs, and design matrix for the new design process
8. Decompose the FR-DP tree for sequential designing
9. Use the newly defined process